

SRI GANGANAGAR, India

WMO: 421230

Lat: 29.917N

Lon: 73.883E

Elev: 581

StdP: 14.39

Time Zone: 5.50 (IND)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 41.2	(c) 43.4	(d) 34.5	(e) 29.8	(f) 61.7	(g) 37.8	(h) 34.0	(i) 58.0	(j) 6.2	(k) 58.1	(l) 5.2	(m) 58.7	(n) 0.3	(o) N/A	(p) 0.356

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	20.0	111.1	75.9	108.3	75.8	105.7	76.3	84.2	96.1	83.6	95.1	83.0	94.3	2.2	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.8	169.0	89.1	81.1	165.0	89.1	80.4	161.4	89.0	49.2	96.3	48.5	95.5	47.8	94.7	86.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 6.7	(b) 5.3	(c) 4.6	(e) 37.3	(f) 115.3	(g) 2.5	(h) 3.3	(i) 35.5	(j) 117.7	(k) 34.0	(l) 119.6	(m) 32.6	(n) 121.5	(o) 30.7	(p) 123.9		
(a) 6.7	(b) 5.3	(c) 4.6	(e) 36.9	(f) 85.5	(g) 2.2	(h) 0.9	(i) 35.3	(j) 86.1	(k) 34.0	(l) 86.6	(m) 32.7	(n) 87.1	(o) 31.1	(p) 87.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	77.4	54.5	61.5	71.7	83.1	91.5	93.8	92.0	90.0	86.4	79.2	67.1	57.4
	DBStd	14.59	4.33	4.74	6.00	5.97	5.20	5.64	4.09	3.46	3.82	4.98	4.09	5.04
	HDD50	17	11	0	0	0	0	0	0	0	0	0	0	6
	HDD65	716	324	119	12	0	0	0	0	0	0	0	23	238
	CDD50	10033	151	322	674	994	1287	1315	1302	1239	1093	906	514	236
	CDD65	5258	0	21	221	544	822	865	837	774	643	441	87	3
	CDH74	77288	2	182	2381	7566	13424	14275	13212	11615	8833	5010	756	32
	CDH80	48408	0	23	986	4596	9247	10048	8784	7191	4906	2490	134	3
(14) Wind	WSAvg	1.0	0.7	1.0	1.1	1.4	1.5	1.7	1.5	1.3	1.0	0.5	0.3	0.4
(15) Precipitation	PrecAvg	13.0	0.3	0.7	0.5	0.5	0.7	1.4	3.0	2.8	2.3	0.3	0.1	0.1
	PrecMax	20.2	0.9	2.2	3.1	1.8	2.6	4.7	6.5	7.4	9.0	3.0	0.6	1.0
	PrecMin	6.1	0.0	0.0	0.0	0.0	0.0	0.2	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	3.9	0.3	0.7	0.8	0.5	0.7	1.3	1.8	2.2	2.5	0.6	0.1	0.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.7	83.1	98.1	108.8	114.6	115.1	109.3	103.5	102.4	99.0	86.9	77.7
		MCWB	59.5	65.1	69.5	70.8	74.0	75.3	81.0	81.0	78.3	73.8	67.6	60.5
	2%	DB	69.6	79.1	93.2	105.1	111.2	111.5	105.1	101.6	99.8	96.0	83.3	74.0
		MCWB	59.0	63.9	68.9	69.9	73.7	77.7	80.7	80.9	78.0	73.7	66.7	60.6
	5%	DB	67.1	75.6	89.0	102.1	108.4	108.8	102.9	99.9	98.1	93.7	80.9	71.3
		MCWB	56.9	62.0	68.6	69.2	73.5	78.2	81.0	80.6	77.9	73.7	66.1	59.5
	10%	DB	64.7	72.4	85.5	98.5	105.7	106.1	100.7	98.0	96.1	91.2	78.4	68.9
		MCWB	55.6	60.3	67.4	69.0	73.6	78.6	81.2	80.9	78.4	72.9	65.2	58.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	62.3	68.5	73.8	75.5	81.1	84.4	85.1	84.9	84.3	80.1	70.4	64.3
		MCDB	68.9	77.6	88.5	99.4	102.1	99.9	96.8	94.7	93.9	92.8	81.3	72.5
	2%	WB	60.0	66.0	71.8	73.0	78.8	83.4	84.3	84.0	83.1	78.3	69.0	62.1
		MCDB	67.4	75.2	87.0	95.3	101.8	98.7	96.9	94.1	92.1	90.0	80.3	71.1
	5%	WB	58.4	64.0	70.3	71.6	77.1	82.5	83.6	83.5	82.0	76.7	67.7	60.4
		MCDB	65.3	72.7	84.5	93.9	100.2	97.8	96.0	93.2	90.6	87.4	78.8	69.8
	10%	WB	56.7	62.0	68.6	70.4	75.8	81.5	82.9	82.8	80.9	75.1	66.2	58.7
		MCDB	62.8	70.1	82.6	92.3	98.6	97.0	95.0	92.3	89.7	86.5	76.3	67.4
(35) Mean Daily Temperature Range	5% DB	MDBR	17.3	20.2	22.9	26.4	25.2	20.0	15.5	13.9	16.6	22.6	21.3	19.9
		MCDBR	21.6	24.6	28.4	30.8	29.0	24.6	17.8	16.0	19.8	24.4	24.6	23.7
		MCWBR	11.6	11.0	9.2	7.7	7.1	6.0	4.5	4.3	4.9	7.0	10.6	12.0
	5% WB	MCDBR	18.8	20.9	25.0	26.3	24.0	18.1	15.9	14.0	14.7	19.2	20.7	21.3
		MCWBR	11.3	10.5	9.9	7.9	7.8	6.1	4.9	4.4	5.2	6.4	10.1	11.7
(40) Clear-Sky Solar Irradiance	taub		0.490	0.476	0.503	0.580	0.668	0.761	0.793	0.686	0.592	0.634	0.608	0.533
	taud		1.838	1.928	1.889	1.751	1.608	1.493	1.474	1.660	1.800	1.618	1.629	1.750
	Ebn at Noon		224	246	249	235	215	195	188	208	223	197	187	201
	Edh at Noon		56	55	61	73	84	94	95	79	66	75	68	58
(44) All-Sky Solar Radiation	RadAvg		963	1335	1751	2050	2192	1994	1807	1794	1727	1475	1124	967
	RadStd		101	97	87	80	58	108	98	108	80	69	71	58

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page